

7(a)	number of protons = 83 and number of neutrons = 129	A1
7(b)	$\lambda = 3.8 \times 10^{-12}$	C1
	$f = 3.0 \times 10^8 / 3.8 \times 10^{-12}$	C1
	$f = 7.9 \times 10^{19} (7.89 \times 10^{19}) \text{ Hz}$	A1
7(c)	use an electric field (at an angle to the beam)	M1
	α is deflected <u>and</u> γ is undeflected	A1
7(d)	<i>either</i>	
	energy = $9.3 \times 10^{-13} / 1.8 \times 10^5 (= 5.17 \times 10^{-18} \text{ J})$	C1
	$= 5.17 \times 10^{-18} / 1.6 \times 10^{-19}$ $= 32 (32.3) \text{ eV}$	A1
	<i>or</i>	
	energy = $9.3 \times 10^{-13} / 1.6 \times 10^{-19} (= 5.81 \times 10^6 \text{ eV})$	(C1)
	$= 5.81 \times 10^6 / 1.8 \times 10^5$ $= 32 (32.3) \text{ eV}$	(A1)